1. PROBLEM STATEMENT

Forests are an essential part of the environment; they provide critical natural habitats, resources, and recreational areas, while absorbing enormous amounts of carbon dioxide. However, forests are increasingly afflicted by drought, pests, and disease. Current methods for determining a tree's health are time consuming and are often commissioned only after visible damage is observed, or they harm the tree during testing. A solution is needed to address this problem that provides a low-cost, reliable, and nondestructive method to continuously monitor tree health throughout forested areas.

1.1. Need Statement

According to the U.S Department of Agriculture, nearly 11% of forests in the United States are exposed to drought, expected to rise to 38% by mid-century, while 11-16% of tree species are being driven to extinction by insects and disease [1], [2]. Low water content can indicate a tree may be suffering from diseases such as vascular wilt, root rot, or drought conditions for otherwise healthy trees. Current methods of monitoring tree health involve foresters manually assessing trees by estimating water content in the leaves or cutting down trees to measure biomass. These methods are often time-consuming or require visible indicators of tree decay, often making the necessary interventions too late for any individual tree or potentially an entire surrounding population [3]. A portable, reliable, and nondestructive device for monitoring tree health is needed to allow forest health to be assessed more efficiently.

1.2. Objective Statement

The objective of this project is to design and build a robot that monitors the health of trees and forests. The robot measures changes in GPS signal attenuation as the signal passes through tree branches and leaves. Because water in vegetation affects the strength of GPS signals, the signal attenuation can be used to estimate the water content in the tree. The Forest Buddy is guided by waypoints set by the user walking through the forest but can be operated through a remote control if necessary. As the Forest Buddy collects data, it sends this data to a database where it can be compared with control data from a base station which is in an open field. Researchers or land agencies can then use this data to estimate the health of forest**s**.

1.3. Background and Related Work

Currently, forest health is assessed using manual inspection, drone networks, and fixed sensor networks. These methods primarily rely on visible symptoms, such as needle discoloration, or directly detecting a pathogen, rather than measuring water content [4], [5]. Manual inspection of forests is impractical for large areas [4], [5]. Remote sensing techniques such as drone and satellite imaging can capture broader spectral data like chlorophyll levels and canopy color, but environmental conditions and coverage limitations can make consistent, large-scale monitoring difficult [4], [5]. Fixed sensor networks can continuously monitor tree health, but require installing a device on every tree, which is prohibitively expensive for large scale areas [4]. Forest Buddy has the same core concepts as a previous project called Leaf Link, which was limited by a pulley system and could not cover many trees. Forest Buddy will be effective at detecting sickness in a time efficient, inexpensive, non-destructive, and scalable way by using water content as an early indicator of sickness rather than relying on visible symptoms alone. Its mobility allows it to collect more data from a larger number of trees, helping researchers monitor forests in a more efficient way.

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